CHEST

Predictors of Mortality and Progression in Scleroderma-Associated Interstitial Lung Disease

A Systematic Review

Tiffany A. Winstone, MD; Deborah Assayag, MD; Pearce G. Wilcox, MD; James V. Dunne, MD; Cameron J. Hague, MD; Jonathon Leipsic, MD; Harold R. Collard, MD, FCCP; and Christopher J. Ryerson, MD

BACKGROUND: Interstitial lung disease (ILD) is the leading cause of morbidity and mortality in patients with systemic sclerosis (SSc); however, prognostication of SSc-associated ILD (SSc-ILD) remains challenging. We conducted a systematic review to identify variables that predict mortality and ILD progression in SSc-ILD.

METHODS: Three databases were searched to identify all studies relating to predictors of mortality or ILD progression in SSc-ILD. Studies were eligible if they were published in English and included ≥ 10 adults with SSc-ILD. Two authors independently reviewed and extracted data from acceptable studies.

RESULTS: The initial search identified 3,145 unique citations. Twenty-seven studies, including six abstracts, met the inclusion criteria. A total of 1,616 patients with SSc-ILD were included. Patient-specific, ILD-specific, and SSc-specific variables predicted mortality and progression; however, most predictors were identified in only one study. Most studies did not fully account for potential confounders, and none of the studies included a validation cohort. Older age, lower FVC, and lower diffusing capacity of carbon monoxide predicted mortality in more than one study. Male sex, extent of disease on high-resolution CT (HRCT) scan, presence of honeycombing, elevated KL-6 values, and increased alveolar epithelial permeability were identified as predictors of both mortality and ILD progression on unadjusted analysis. The extent of disease on HRCT scan was the only variable that independently predicted both mortality and ILD progression.

CONCLUSIONS: Mortality and ILD progression were predicted by several patient-specific, ILD-specific, and SSc-specific factors. Additional prospective studies are required to validate these preliminary findings and to identify combinations of variables that accurately predict the prognosis of SSc-ILD.

CHEST 2014; 146(2):422-436

Manuscript received November 5, 2013; revision accepted February 1, 2014; originally published Online First February 27, 2014.

ABBREVIATIONS: DLCO = diffusing capacity of the lung for carbon monoxide; HRCT = high-resolution CT; ILD = interstitial lung disease; IPF = idiopathic pulmonary fibrosis; SSc = systemic sclerosis; SSc-ILD = systemic sclerosis-associated interstitial lung disease

AFFILIATIONS: From the Department of Medicine (Drs Winstone, Wilcox, Dunne, and Ryerson) and the Department of Radiology (Drs Hague and Leipsic), University of British Columbia, Vancouver, BC, Canada; and the Department of Medicine (Drs Assayag and Collard), University of California San Francisco, San Francisco, CA.

FUNDING/SUPPORT: The authors have reported to *CHEST* that no funding was received for this study.

CORRESPONDENCE TO: Christopher J. Ryerson, MD, Department of Medicine, University of British Columbia-St. Paul's Hospital, 1081 Burrard St, Ward 8B, Vancouver, BC, V6Z 1Y6, Canada; e-mail: chris.ryerson@hli.ubc.ca

© **2014 AMERICAN COLLEGE OF CHEST PHYSICIANS.** Reproduction of this article is prohibited without written permission from the American College of Chest Physicians. See online for more details.

DOI: 10.1378/chest.13-2626

Pulmonary involvement is a prominent feature in systemic sclerosis (SSc) and occurs more frequently in SSc than in any other connective tissue disease.[1] The two most frequent types of lung involvement are interstitial lung disease (ILD) and pulmonary arterial hypertension.[2] The reported prevalence of ILD in patients with SSc ranges from 25% to 90%, depending on the subtype of SSc and the criteria used to define ILD.[3]

ILD is the leading cause of mortality in patients with SSc[4]; however, there is substantial variability in the disease course, and prognostication of SSc-associated ILD (SSc-ILD) is challenging. We conducted a systematic review to identify variables that predict mortality and ILD progression in patients with SSc-ILD. Our objectives were to identify variables or composite scores that could be used to help guide management of SSc-ILD and to identify priorities for future research in the prognostication of these patients.

Materials and Methods

Data Sources and Search

We performed a systematic search of the literature using the MEDLINE, Embase, and Evidence-Based Medicine Reviews databases. The search was designed to capture all studies related to predictors of mortality and ILD progression in SSc-ILD. Two authors (T. A. W. and C. J. R.) reviewed citations independently using predefined criteria. Searches covered the period from the onset of the database to April 2013. Bibliographies from selected articles and major reviews were screened for additional relevant publications that were not identified by the database search. Details of the search strategy are available in e-Table 1.

Study Selection

Two authors (T. A. W. and C. J. R.) independently evaluated each study for eligibility, sequentially reviewing the title, abstract, and full publication. Studies were eligible if they were published in English and evaluated potential predictors of survival and/or ILD progression in ≥10 adults with SSc-ILD. ILD progression was defined by worsening FVC or radiologic findings of fibrosis. Studies focusing on treatment effects were excluded based on our objective of identifying variables that could help guide the management of SSc-ILD. There was no restriction on study design, study quality, or duration of follow-up. Published conference abstracts were included if sufficient information was provided to satisfy the inclusion criteria. Differences were resolved by iteration and consensus.

Data Extraction and Quality Assessment

Data were extracted from selected studies using predefined data-abstraction forms that included information on study design, subject characteristics, and study results as they pertained to the prespecified end points. Study quality was assessed using the Critical Appraisal Skills Program (http://www.casp-uk.net). Quality assessment was based on the method of population sampling, the definitions of SSc and ILD, how the outcome was ascertained, and whether confounders were appropriately identified and accounted for in the statistical analysis.

Data Synthesis and Analysis

We report hazard ratios and/or ORs for predictors of mortality and ILD progression in SSc-ILD. Results were grouped according to predefined categories including patient-specific variables (eg, age and sex), ILD-specific variables (eg, measures of ILD severity or activity), and SSc-specific variables (eg, serum autoantibodies and extrapulmonary features of SSc). We qualitatively summarized findings if similar variables were evaluated in multiple studies; however, a formal meta-analysis of results was not possible because of the heterogeneity of the included studies and the inconsistent methods for reporting study results.

Results

Search Results and Study Characteristics

The initial search identified 3,145 unique citations, and 167 were reviewed in full text. Twenty-seven descriptive studies met the eligibility criteria,[5-31] including six abstracts (Fig 1, Table 1).[7,9,21,24,30,31] A total of 1,616 patients with SSc-ILD were included (range, 18-215 patients per study). The duration of follow-up was reported in 21 studies and ranged from 1 to 14 years. There was one prospective cohort study[22] and one retrospective case-control study.[6] All other full-text publications were either retrospective studies or post hoc analyses of prospectively recruited cohorts, including three post hoc analyses[17,23,26] of the Scleroderma Lung Study.[32]

Study quality is summarized in e-Table 2. American College of Rheumatology diagnostic criteria for SSc were used in 17 studies. Eight studies did not specify how the diagnosis of SSc was made.[7,9,15,16,21,25,30,31] Variable criteria were used for the diagnosis of ILD, including the findings of chest high-resolution CT (HRCT) scans,[10,11,14-16,19,21,22,28] chest radiograph,[31] lung biopsy,[8,13] pulmonary function testing,[25] or a combination of these variables.[5,6,12,17,18,20,23,24,26,27] Four studies did not specify the criteria used for ILD diagnosis.[7,9,29,30]

Subject Characteristics

Subject characteristics are summarized in e-Table 3. Mean age ranged from 46 to 55.6 years, 23% of the subjects were men, and 33% had ever smoked in the 15 studies reporting smoking status. The duration of SSc ranged from 1.2 years to 10.5 years in reported studies. ILD severity was variable among studies, as indicated by the wide range of physiologic and radiologic abnormalities.

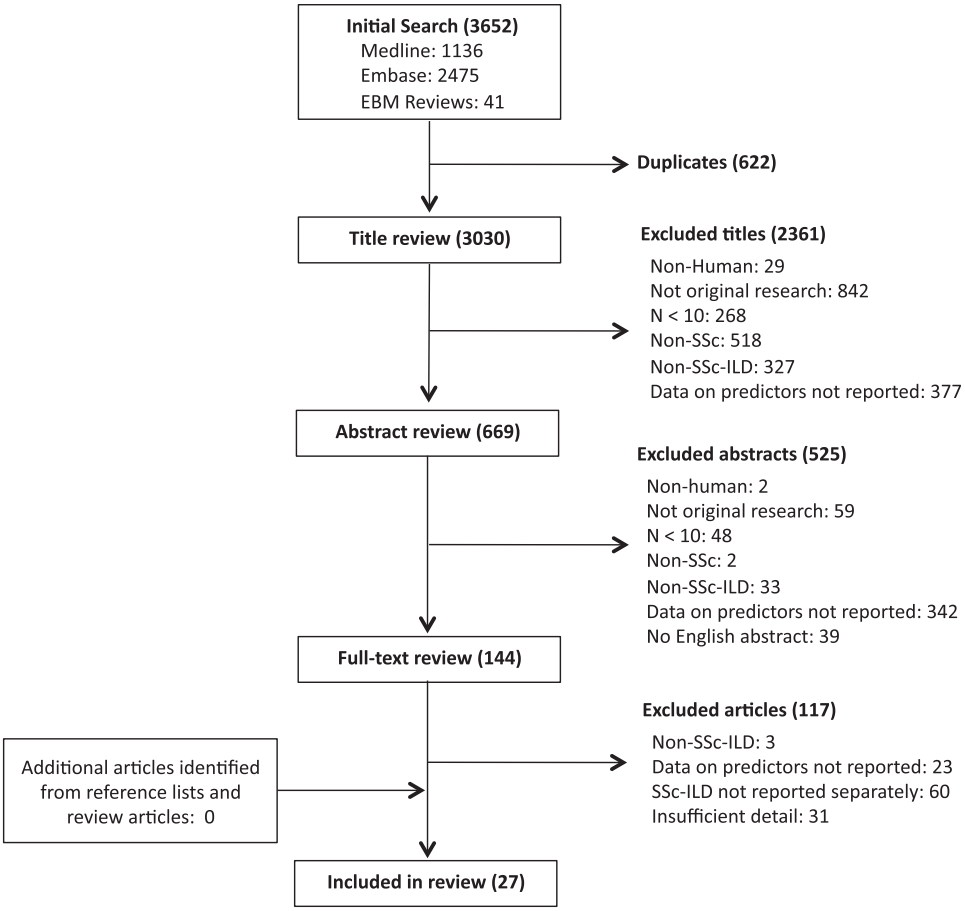

Figure 1 – *Study selection. EBM = evidence-based medicine; SSc = systemic sclerosis; SSc-ILD = systemic sclerosis-associated interstitial lung disease.*

Predictors of Mortality in SSc-ILD

Fourteen studies evaluated predictors of mortality, including a total of 1,198 patients with SSc-ILD (e-Table 4). Several patient-specific, ILD-specific, and SSc-specific factors predicted mortality in SSc-ILD on unadjusted analysis (Fig 2, Table 2). Age, FVC, diffusing capacity of the lung for carbon monoxide (DLCO), HRCT scan fibrosis severity, BAL neutrophilia, and pulmonary hypertension (echocardiography [n = 1] and undefined criteria [n = 1]) were evaluated in multiple studies and were significant predictors in the majority of studies that evaluated these variables. One of five studies examining sex showed a statistically significant association with unadjusted mortality.[15] Change in DLCO over 3 years was a significant predictor of mortality in the single study that evaluated this variable; however, change in FVC over time was not.[8] The presence of anticentromere antibodies predicted unadjusted mortality in only one study.[21] Compared with limited SSc, diffuse SSc did not predict mortality in the four studies that evaluated this possibility.[5,8,11,21]

Seven studies included multivariate models for the prediction of mortality in SSc-ILD (Fig 2, Table 3). The factors adjusted for in each study varied substantially, and details of the multivariate model were not provided in some studies. None of the studies provided a description of model discrimination or calibration. Baseline DLCO was identified as an independent predictor of mortality in four of seven multivariate models.[8,14,15,19] An additional study found that change in DLCO after 3 years also predicted mortality after adjustment for baseline pulmonary function and BAL eosinophil levels.[8] HRCT scan fibrosis severity and pulmonary hypertension were also identified as independent predictors of mortality in more than one study.[14,15] Several additional variables, including increased age, lower oxygen saturation, presence of anti-U11/12 ribonucleoprotein, BAL neutrophil count > 4%, BAL eosinophil count > 2%, and presence of a pericardial effusion, were independent predictors of mortality in only one study.

TABLE 1] Study Characteristics

Study/Year	Enrollment, y	Objective(s)	Recruitment Site (Country)	SSc Diagnostic Criteria	ILD Diagnostic Criteria	Other Key Inclusion Criteria	No. With SSc-ILD (Total No.)	Follow-up
Al-Dhaher et al[5]/2010	1994-2004	Describe morbidity and mortality of SSc, and compare features between those living and deceased	Rheumatology clinic (Canada)	ACR	PFT, CXR, and/or HRCT scan	…	27 (185)	9.1 y[a]
Beretta et al[6]/2007	N/R	Determine if the T-889C IL-1α promoter gene polymorphism affects response to cyclophosphamide in SSc-ILD	N/R (Italy)	ACR	Change in lung function and HRCT scan	Recent deterioration, treated with oral steroids and cyclophosphamide	18 (18)	12 mo
Boin et al[7, b]/2010	N/R	Determine if an abnormal distribution of polarized T-cell subsets in the periphery of patients with SSc is associated with active ILD	N/R (N/R)	N/R	N/R	Undergoing bronchoscopy for suspected active ILD	22 (N/R)	N/R
Bouros et al[8]/2002	1985-1997	Determine histopathologic, physiologic, and BAL predictors of mortality and progression in SSc-ILD	ILD clinic (United Kingdom)	ACR	Biopsy	…	74 (476)	75 mo[c]
De Lauretis et al[9, b]/2010	N/R	Determine whether serum cytokines predict disease progression and mortality in IPF and SSc-ILD	N/R (N/R)	N/R	N/R	…	74 (152)	N/R
De Santis et al[10]/2011	2007-2009	Determine whether BAL B-thymosins predict progression of SSc-ILD	Rheumatology clinic (Italy)	ACR	HRCT scan	Undergoing bronchoscopy with BAL	46 (61)	12 mo
De Santis et al[11]/2012	2003-2005	Determine BAL characteristics and clinical and radiologic parameters associated with SSc-ILD progression and mortality	Rheumatology clinic (Italy)	ACR	HRCT scan	…	73 (110)	1-3 y

(Continued)

TABLE 1] *(continued)*

Study/Year	Enrollment, y	Objective(s)	Recruitment Site (Country)	SSc Diagnostic Criteria	ILD Diagnostic Criteria	Other Key Inclusion Criteria	No. With SSc-ILD (Total No.)	Follow-up
Fertig et al[12]/2009	1982-2005	Describe the clinical features and disease course of U11/U12 RNP-positive and anti-U11/U12 RNP-negative patients with SSc	Rheumatology clinic (United States)	Physician-confirmed diagnosis	CXR or HRCT scan	…	97 (480)	6.9-9.7 y[c]
Fischer et al[13]/2008	1983-2005	Evaluate clinical features and predictors of survival in SSc-ILD	ILD clinic (United States)	ACR	Lung biopsy	UIP or NSIP on histopathology	22 (27)	N/R
Goh et al[14]/2007	1990-1999	Evaluate the prognostic value of BAL cellular profiles in SSc-ILD	Hospital ward (United Kingdom)	ACR	HRCT scan	No overlap connective tissue disorders or current smokers	141 (330)	95 mo[c]
Goh et al[15]/2008	1990-1999	Identify baseline PFT and HRCT scan variables that predict survival in SSc-ILD	Hospital ward (United Kingdom)	N/R	HRCT scan	…	215 (330)	89 mo[c]
Goh et al[16]/2011	1990-1999	Examine the relationship between the rapidity of ^{99m}Tc-DTPA clearance and outcome in IPF and SSc-ILD	N/R (N/R)	N/R	HRCT scan	No current smokers	168 (468)	N/R
Khanna et al[17, d]/2011	2000-2004	Evaluate predictors of worsening FVC % in SSc-ILD	SLS (United States)	ACR	HRCT scan and/or BAL alveolitis	SLS placebo subjects	77 (158)	12 mo[a]
Kim et al[18]/2001	1990-1999	Assess serial HRCT scan findings and PFT parameters in patients with progressive SSc-ILD	Hospital wards (N/R)	ACR	HRCT scan and biopsy	Follow-up CT scan performed at least 5 mo after hospital admission	40 (197)	39 mo[a]
Kuwana et al[31, b]/2012	1980-1995	Evaluate predictors for the development of ESLD in untreated SSc-ILD	N/R (N/R)	N/R	CXR	No immunomodulatory therapy, disease duration < 3 y at diagnosis	50 (N/R)	174 mo[a]
Launay et al[19]/2006	1991-2001	Investigate the CT scan pattern of patients with SSc and identify predictors of radiographic progression	N/R (France)	ACR	HRCT scan	Two HRCT scans separated by ≥ 2 y	50 (156)	5.2 y[a]
Launay et al[20]/2011	2001-2006	Assess survival in SSc-ILD and coexisting PH	ILD clinics (France)	ACR	HRCT scan and PFT	PAH or PH	47 (97)	36 mo after PH diagnosis[a]

(Continued)

TABLE 1] *(continued)*

Study/Year	Enrollment, y	Objective(s)	Recruitment Site (Country)	SSc Diagnostic Criteria	ILD Diagnostic Criteria	Other Key Inclusion Criteria	No. With SSc-ILD (Total No.)	Follow-up
Le Gouellec et al[21, b]/2012	N/R	Assess the prognostic value of baseline clinical characteristics, pulmonary function, and HRCT scan findings in SSc-ILD	N/R (France)	N/R	HRCT scan	…	75 (N/R)	N/R
Lopes et al[22]/2011	2002-2005	Evaluate whether HRCT scan honeycombing predicts worsening FVC or D_{LCO}	N/R (Brazil)	Clinical and biochemical (N/R)	HRCT scan	No current smokers	35 (63)	61 mo[a]
Roth et al[23, d]/2011	2000-2004	Identify baseline predictors of response to cyclophosphamide in SSc-ILD	SLS (United States)	ACR	HRCT scan and/or BAL alveolitis	SLS subjects	112 (158)	18 mo
Shahane et al[24, b]/2010	N/R	Determine the effect of coexistent PH on progression of SSc-ILD	ILD clinic (United States)	ACR	HRCT scan and PFT	…	133 (N/R)	6 y
Steen et al[25]/1994	1972-1989	Identify risk factors for severe or progressive SSc-ILD	Rheumatology clinic (United States)	N/R	PFT	$FEV_1/FVC > 70\%$	116 (1,259)	N/R
Strange et al[26, d]/2008	2000-2004	Evaluate whether BAL cellularity can identify a distinct phenotype, predict rate of progression, and/or predict response to cyclophosphamide in SSc-ILD	SLS (United States)	ACR	HRCT scan and/or BAL alveolitis	SLS placebo subjects	141 (267)	12 mo
Swigris et al[27]/2009	1983-2005	Determine the association of Spo_2 and Sao_2 and whether exertional Spo_2 predicts mortality in SSc-ILD	ILD clinic (United States)	ACR	CXR, HRCT scan, or biopsy	No PH	83 (83)	7.1 y[c]
Trad et al[28]/2006	1980-2004	Determine whether PAH is a risk factor for mortality in dSSc	Hospital ward (France)	ACR	HRCT scan	…	52 (86)	104 mo[a]

(Continued)

TABLE 1 *(continued)*

Study/Year	Enrollment, y	Objective(s)	Recruitment Site (Country)	SSc Diagnostic Criteria	ILD Diagnostic Criteria	Other Key Inclusion Criteria	No. With SSc-ILD (Total No.)	Follow-up
Volpinari et al[29]/2011	N/R	Evaluate the role and the prognostic value of BAL in patients with SSc-ILD	N/R (N/R)	ACR	N/R	No PH, negative cultures on BAL	79 (154)	18-36 mo[e]
Zhang et al[30, b]/2012	N/R	Determine the association between gastroesophageal involvement and progression of SSc-ILD	Multicenter database (Canada)	N/R	N/R	...	N/R (1,043)	N/R

ACR = American College of Rheumatology; CXR = chest radiograph; DLco = diffusing capacity of the lung for carbon monoxide; dSSc = diffuse systemic sclerosis; DTPA = diethylene thiamine pentacetate clearance; ESLD = end-stage lung disease; HRCT = high-resolution CT; ILD = interstitial lung disease; IPF = idiopathic pulmonary fibrosis; N/R = not reported; NSIP = nonspecific interstitial pneumonia; PAH = pulmonary arterial hypertension; PFT = pulmonary function test; PH = pulmonary hypertension; RNP = ribonucleoprotein; Sao_2 = arterial oxygen saturation; SLS = Scleroderma Lung Study; Spo_2 = oxygen saturation; SSc = systemic sclerosis; SSc-ILD = systemic sclerosis-associated interstitial lung disease; UIP = usual interstitial pneumonia.

[a]Mean.

[b]Published in abstract form only.

[c]Median.

[d]SLS post hoc analysis.

[e]Range.

Predictors of ILD Progression in SSc-ILD

Twenty studies, which included a total of 1,524 patients with SSc-ILD, evaluated predictors of ILD progression. Several patient-specific, ILD-specific, and SSc-specific factors were identified as predictors of ILD progression on unadjusted analysis (e-Table 5). HRCT scan fibrosis severity and a shorter duration of SSc were the only significant predictors of ILD progression in more than one study.[17,22-25]

Four studies conducted multivariate analyses that evaluated potential independent predictors of ILD progression[16,23,24,26]; however significant predictors of progression were identified in only two of these studies (Table 3).[16,24] HRCT scan fibrosis severity predicted change in FVC in one study with adjustment for the presence of pulmonary hypertension.[24] Abnormal alveolar epithelial permeability (determined by abnormal or fast pulmonary clearance of diethylene thiamine pentacetate) predicted time to FVC decline in the single study that evaluated this variable, including with adjustment for disease severity by HRCT scan or physiologic measures.[16]

Discussion

We conducted a rigorous systematic review of the literature to identify variables that predict mortality or ILD progression in patients with SSc-ILD. Several patient-specific, ILD-specific, and SSc-specific variables predicted mortality and ILD progression in SSc-ILD; however, included studies were of variable methodologic quality, and the predictors identified in this review should be considered within the context of these limited data. A minority of studies performed a multivariate analysis, and these studies did not identify a variable or combination of variables that accurately predicted prognosis of SSc-ILD. The paucity of data highlights the challenge of accurately identifying prognostic factors in SSc-ILD and provides justification for additional research.

FVC is the only pulmonary function measurement that has been used as a primary outcome in randomized controlled trials in SSc-ILD.[32] Although baseline FVC was an unadjusted predictor of mortality in multiple studies,[8,15,21,25] it was not identified as an independent predictor in any of the multivariate analyses. Change in FVC was also not predictive of mortality[8]; however, this was evaluated in only a single study that was likely underpowered to test this hypothesis. The inconsistent association of FVC and change in FVC with mortality in previous studies suggests that further research is required

TABLE 2] Statistically Significant Predictors of Mortality in SSc-ILD on Unadjusted Analysis

Predictor Variable and Study/Year	Outcome	Hazard Ratio	95% CI	*P* Value
Patient-specific variables				
Sex				
Goh et al[15]/2008	Mortality (male)	1.98	1.20-3.26	<.01
Age				
Goh et al[15]/2008	Mortality	1.03	1.01-1.05	.001
Goh et al[15]/2008	Mortality (all patients)	1.03	1.01-1.05	<.005
Age at ILD diagnosis				
Le Gouellec et al[21, a]/2012	Survival	N/R	N/R	<.05
Age at symptom onset				
Fertig et al[12]/2009	Mortality	1.06	1.04-1.09	<.0001
ILD-specific variables				
Physiologic variables				
FVC				
Le Gouellec et al[21, a]/2012	Survival	N/R	N/R	<.05
Bouros et al[8]/2002	Mortality (all patients)	N/R	N/R	.007
Goh et al[15]/2008	Mortality	0.99	0.97-0.99	<.01
Steen et al[25]/1994	10-y survival	N/R	N/R	<.01
FVC %<60				
Goh et al[15]/2008	Mortality	1.92	1.14-3.23	.01
FVC %<65				
Goh et al[15]/2008	Mortality	1.78	1.11-2.85	.02
FVC %<70				
Goh et al[15]/2008	Mortality	2.11	1.34-3.32	.001
FVC %<75				
Goh et al[15]/2008	Mortality	1.93	1.22-3.05	.005
D_{LCO}				
Le Gouellec et al[21, a]/2012	Survival	N/R	N/R	<.05
Bouros et al[8]/2002	Mortality (all patients)	N/R	N/R	.004
Bouros et al[8]/2002	Mortality (patients with NSIP)	N/R	N/R	.04
Goh et al[15]/2008	Mortality	0.95	0.94-0.97	<.0005
Launay et al[20]/2011	Mortality	0.96	N/R	.01
Trad et al[28]/2006	Deceased at study completion (mean 8.7 y)	N/R	N/R	.007
D_{LCO}<30% (vs ≥30%)				
Launay et al[20]/2011	Survival at 36 mo	N/R	N/R	.004
Change in D_{LCO} at 3 y				
Bouros et al[8]/2002	Mortality (patients with NSIP)	N/R	N/R	.002
FVC to D_{LCO} ratio				
Launay et al[20]/2011	Survival[b]	1.004	1.001-1.007	.003
FEV_1				
Goh et al[15]/2008	Mortality	0.98	0.97-0.99	<.01

(Continued)

TABLE 2] *(continued)*

Predictor Variable and Study/Year	Outcome	Hazard Ratio	95% CI	*P* Value
Spo_2max < 89% (vs ≥ 89%)				
Swigris et al[27]/2009	Mortality	2.4	1.2-4.9	.02
Spo_2max fell > 4 points from baseline vs remained within 4 points				
Swigris et al[27]/2009	Mortality	2.4	1.1-5.0	.02
Difference between baseline Spo_2 and Spo_2max				
Swigris et al[27]/2009	Mortality	1.08	1.03-1.14	.002
Presence of ESLD				
Kuwana et al[31, a]/2012	Survival	N/R	N/R	< .0001
Radiologic variables				
Alveolar score				
De Santis et al[11]/2012	Mortality	N/R	N/R	.035
Honeycombing				
De Santis et al[11]/2012	Mortality	N/R	N/R	.014
De Santis et al[11]/2012	Respiratory mortality	1.23	1.10-1.44	< .05
Bilateral honeycombing				
De Santis et al[11]/2012	Mortality	N/R	N/R	.045
Extent of disease				
Goh et al[15]/2008	Mortality	1.03	1.01-1.04	< .0005
> 15%				
Goh et al[15]/2008	Mortality	2.05	1.30-3.23	.002
> 20%				
Goh et al[15]/2008	Mortality	2.48	1.57-3.92	< .0005
> 25%				
Goh et al[15]/2008	Mortality	2.31	1.43-3.72	.001
> 30%				
Goh et al[15]/2008	Mortality	2.86	1.72-4.76	< .0005
> 35%				
Goh et al[15]/2008	Mortality	2.80	1.53-5.15	.001
> 40%				
Goh et al[15]/2008	Mortality	2.79	1.38-5.64	.004
Extent of reticular pattern				
Goh et al[15]/2008	Mortality	1.05	1.03-1.07	< .0005
Proportion of ground glass				
Goh et al[15]/2008	Mortality	0.99	0.98-0.99	.001
Coarseness of reticulation				
Goh et al[15]/2008	Mortality	1.15	1.06-1.25	.001
Bronchoscopic variables				
% neutrophil				
De Santis et al[11]/2012	Mortality	N/R	N/R	.007
Eosinophil (> 2%)				

(Continued)

TABLE 2] *(continued)*

Predictor Variable and Study/Year	Outcome	Hazard Ratio	95% CI	*P* Value
Bouros et al[8]/2002	Mortality (patients with NSIP)	N/R	N/R	.03
Neutrophil (>4%)				
Goh et al[14]/2007	Mortality	2.23	1.20-4.14	.01
Laboratory variables				
C-reactive protein				
Le Gouellec et al[21, a]/2012	Survival	N/R	N/R	<.05
Hemoglobin level				
Le Gouellec et al[21, a]/2012	Survival	N/R	N/R	<.05
Normal vs elevated KL-6				
Kuwana et al[31, a]/2012	Survival	N/R	N/R	<.0001
Other variables - DTPA				
Increasing rapid total clearance, min				
Goh et al[16]/2011	Mortality	1.02	1.00-1.03	.02
SSc-related variables				
Anticentromere antibody				
Le Gouellec et al[21, a]/2012	Survival	N/R	N/R	<.05
Diffuse systemic sclerosis				
De Santis et al[11]/2012	Mortality	10.7	2.1-56.0	<.05
PAH/PH				
De Santis et al[11]/2012	Mortality	N/R	N/R	.048
Goh et al[15]/2008	Mortality	4.78	2.68-8.54	<.0005
Goh et al[15]/2008	Mortality	2.26	1.18-4.34	.01
Pericardial effusion				
Launay et al[20]/2011	Mortality	2.44	N/R	.04
LVEF				
Le Gouellec et al[21, a]/2012	Survival	N/R	N/R	<.05

LVEF = left ventricular ejection fraction; Spo_2max = Spo_2 at maximal exercise. See Table 1 legend for expansion of other abbreviations.
[a]Published in abstract form only.
[b]Hazard ratio > 1.0 indicates improved survival with higher FVC to DLCO ratio.

to validate FVC as an outcome measure in SSc-ILD. The paucity of data limits speculation about the prognostic ability of FVC; however, these findings suggest that factors unrelated to ILD severity are important in determining prognosis in SSc-ILD. Conversely, DLCO was identified in multiple studies as a significant predictor of mortality in both unadjusted and adjusted analyses. The apparently stronger association of DLCO with mortality is likely related to DLCO being a marker for the severity of both ILD and pulmonary hypertension, two major contributors to mortality in SSc.[4]

The ability of DLCO to capture the severity of both fibrosis and pulmonary hypertension may also account for the lack of independent association between FVC and mortality in SSc-ILD, because FVC represents only one of these contributors to mortality. In addition, confounders may weaken the association of FVC with ILD severity in SSc (eg, extrathoracic restriction from extensive skin disease and respiratory muscle weakness). These findings differ from those of previous studies in idiopathic pulmonary fibrosis (IPF) in which FVC is a more reliable predictor of mortality,[33-35] likely related to the less frequent association of IPF with pulmonary arterial hypertension. Additional studies are required to confirm the prognostic ability of the FVC to DLCO ratio, a measure that may provide a more accurate estimate of pulmonary arterial hypertension.

Radiologic variables were significant predictors of prognosis in several studies[17,22,24]; however, the extent of disease on HRCT scan was the only predictor of mortality on adjusted analysis.[15] The radiologic and histopathologic features of the usual interstitial pneumonia

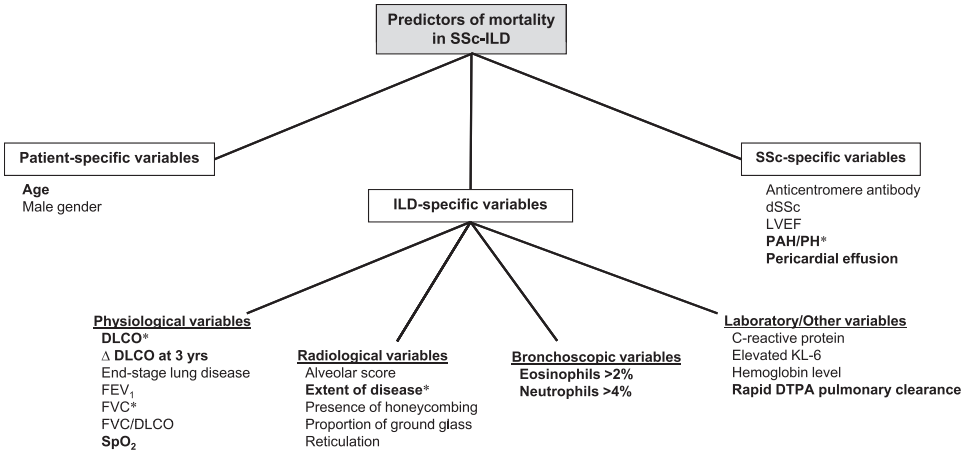

Figure 2 – *Statistically significant predictors of mortality in SSc-ILD. Variables listed in bold were associated with mortality on both bivariate and multivariate analysis. *Statistically significant predictors of mortality that were identified in multiple studies. DLCO = diffusing capacity of the lung for carbon monoxide; dSSc = diffuse SSc; DTPA = diethylene thiamine pentacetate clearance; ILD = interstitial lung disease; LVEF = left ventricular ejection fraction; PAH = pulmonary arterial hypertension; PH = pulmonary hypertension; SpO_2 = oxygen saturation. See Figure 1 legend for expansion of other abbreviations.*

pattern had borderline prognostic ability in SSc-ILD[8,11,13,15]; however, only two of these studies reported histopathologic features, and these limited data prohibit definitive conclusions.[8,13] These findings differ from those of IPF and other fibrotic ILDs, in which the radiologic and histopathologic features of usual interstitial pneumonia are strong predictors of poor outcome.[36-38] Although ground-glass opacification is generally considered a more inflammatory and potentially more treatable and reversible process than denser areas of fibrosis, the proportion of ground-glass opacification was not an independent predictor of progression on multivariate analysis in the single study that evaluated this variable.[15] Additional research is required to determine the relative prognostic importance of ILD pattern (eg, usual interstitial pneumonia vs nonspecific interstitial pneumonia) and severity (ie, radiologic fibrosis score) in SSc-ILD.

A shorter duration of SSc predicted physiologic worsening in two studies; however, disease duration was not an independent predictor of progression on multivariate analysis.[4,23] The apparent rapid progression of SSc-ILD early in disease may be accounted for by survivorship bias. Specifically, patients with rapid progression likely experience frequent early mortality and are not included in the group of patients who survived to have longer-term follow-up. An unadjusted comparison of patients with different disease durations will, therefore, falsely suggest that patients progress more rapidly early during the course of their disease. More rigorous evaluation of this potential association is required, focusing on the within-patient rate of progression in early vs late disease.

Previous studies have not identified a variable or combination of variables that can accurately predict the prognosis of SSc-ILD. Only one study reported some performance characteristics of a prognostic variable, showing that a DLCO < 30% had a sensitivity of 55% and specificity of 83% for death during a variable follow-up period.[20] Other studies have not reported model performance, and no studies have been specifically designed to derive a clinical prediction model to estimate prognosis in SSc-ILD. Additional studies are, therefore, required to derive and validate clinical prediction models for SSc-ILD and to test other models that have been developed in similar entities such as IPF.[33-35] An accurate clinical prediction model could have a significant impact on patient care in SSc-ILD by identifying high-risk individuals who may benefit from closer surveillance, more aggressive therapy with potentially toxic medications, or earlier referral for lung transplant.

The main limitation of this systematic review is that the included studies were of variable methodologic quality. Most predictors were identified in only one study and few studies included rigorous multivariate analyses that attempted to adjust for potential confounders. In addition, there was potential for publication bias; however, this possibility is mitigated by our extensive search of multiple databases, including conference abstracts, as well as our primary interest in positive findings, which are more likely to be published. Finally, the

TABLE 3] Significant Predictors of Mortality and Progression in SSc-ILD on Multivariate Analysis

Predictor Variable and Study/Year	Outcome	Hazard Ratio	95% CI	*P* Value	Variables in Model
Patient-specific variables					
Age at diagnosis					
Trad et al[28]/2006	Mortality	1.073	1.003-1.149	.042	PAH, cancer, heart and kidney involvement
ILD-specific variables					
Physiologic variables					
D_{LCO}					
Bouros et al[8]/2002	Mortality (patients with NSIP)	N/R	N/R	.04	BAL % eosinophils
Goh et al[14]/2007	Mortality	0.97	0.94-0.99	.004	BAL neutrophilia and PAH
Goh et al[15]/2008	Mortality	N/R	N/R	<.0005	Age, sex, smoking status, proportion of ground glass, coarseness of reticulation
Launay et al[20]/2011	Survival	3.96	1.56-10.06	.004	N/R
Change in D_{LCO} at 3 y					
Bouros et al[8]/2002	Mortality (patients with NSIP)	N/R	N/R	<.05	Baseline FVC, D_{LCO}, BAL % eosinophils
Difference between baseline Sp_{O_2} and Sp_{O_2}max					
Swigris et al[27]/2009	Mortality	1.07	1.01-1.14	.02	FVC %
Radiologic variables					
Extent of disease					
Goh et al[15]/2008	Mortality	N/R	N/R	<.0005	Age, sex, smoking status, proportion of ground glass, coarseness of reticulation
Extent of reticular pattern					
Goh et al[15]/2008	Mortality	N/R	N/R	<.001	Age, sex, smoking status, proportion of ground glass, coarseness of reticulation
Extensive vs limited disease					
Goh et al[15]/2008	Mortality	3.66	2.25-5.97	<.0005	Age, sex, smoking status, proportion of ground glass, coarseness of reticulation
Shahane et al[24, a]/2010	Rate of FVC decline	N/R	N/R	<.001	PH
Bronchoscopic variables					
Neutrophil count (>4%)					
Goh et al[14]/2007	Mortality	2.02	1.07-3.82	.03	Extent of disease
Goh et al[14]/2007	2-y mortality	N/R	N/R	.01	Extent of disease on HRCT scan
Goh et al[14]/2007	2-y mortality	N/R	N/R	.02	D_{LCO}
Goh et al[14]/2007	2-y mortality	N/R	N/R	.02	PH
Eosinophil count (>2%)					
Bouros et al[8]/2002	Mortality (patients with NSIP)	N/R	N/R	.02	D_{LCO}
Laboratory variables					
Anti-U11/12 RNP-positive					

(Continued)

TABLE 3] *(continued)*

Predictor Variable and Study/Year	Outcome	Hazard Ratio	95% CI	*P* Value	Variables in Model
Fertig et al[12]/2009	Mortality	2.25	1.20-4.24	.01	Age of symptom onset, sex, tobacco use
Other variables - DTPA					
Increasing rapid total clearance					
Goh et al[16]/2011	Time to FVC decline	N/R	N/R	<.05	Severity of disease
Presence of abnormal clearance					
Goh et al[16]/2011	Time to FVC decline	N/R	N/R	<.05	Severity of disease
Increasing rapid fast clearance, min					
Goh et al[16]/2011	Time to FVC decline	1.15	1.03-1.28	<.01	Age, sex, smoking status, disease severity (CPI)
Goh et al[16]/2011	Time to FVC decline	1.15	1.03-1.27	.01	Age, sex, smoking status, disease severity (HRCT scan)
% fast component					
Goh et al[16]/2011	Time to FVC decline	1.02	1.00-1.04	.04	Age, sex, smoking status, disease severity (CPI)
Goh et al[16]/2011	Time to FVC decline	1.02	1.00-1.04	.04	Age, sex, smoking status and disease severity (HRCT scan)
SSc-specific variables					
PAH/PH					
Goh et al[14]/2007	Mortality	2.73	1.29-5.79	.009	BAL neutrophilia and DLCO
Goh et al[15]/2008	Mortality	N/R	N/R	<.0005	Age, sex, smoking status, proportion of ground glass, coarseness of reticulation
Trad et al[28]/2006	Mortality	5.07	1.09-23.8	.038	Cancer, heart and kidney involvement, age at diagnosis
Pericardial effusion					
Launay et al[20]/2011	Survival	0.95	0.92-0.98	.002	N/R

CPI = composite physiologic index. See Table 1 and 2 legends for expansion of other abbreviations.
[a]Published in abstract form only.

heterogeneity of the included studies prevented pooling of study results, and it is, thus, more challenging to synthesize data from these sources. For example, the variable methods for assessing radiologic ILD severity may account for some of the inconsistencies across studies. Despite these limitations, the data summarized in this review provide a framework for future research in this field and suggest several areas for further investigation. In particular, few studies have evaluated SSc-specific variables (eg, autoantibodies and extrapulmonary manifestations) and these topics require further study.

Conclusions

In summary, we found that several patient-specific, ILD-specific, and SSc-specific features predict worse prognosis in SSc-ILD. DLCO was the most consistent predictor of mortality and may help identify patients with a poor prognosis; however, more rigorous studies are needed to confirm and expand on these preliminary findings. A clearer understanding of how to determine prognosis with variables such as DLCO is required prior to incorporation of these findings into clinical practice. Additional research is specifically required to derive and validate clinical prediction tools in SSc-ILD and to test

models developed in other ILD subtypes. Such studies would likely require a coordinated multicenter effort given the relative rarity of SSc-ILD and the need for a large and distinct cohort for appropriate validation.

Although such a study would present many challenges, there is a significant unmet clinical need for an accurate clinical prediction model in SSc-ILD, and validation of such a model should be a priority for future studies.

Acknowledgments

Author contributions: C. J. R. takes responsibility as the guarantor of the manuscript, including the data and analysis. T. A. W. and C. J. R. contributed to the design of the study, primary analysis and interpretation of the data, production of the initial draft of the manuscript, and approval of the final version and D. A., P. G. W., J. V. D., C. J. H., J. L., and H. R. C. contributed to the analysis and interpretation of the data, drafting of the manuscript, and approval of the final version.

Financial/nonfinancial disclosures: The authors have reported to *CHEST* the following conflicts of interest: Dr Collard is a consultant for Intermune. Drs Winstone, Assayag, Wilcox, Dunne, Hague, Leipsic, and Ryerson have reported that no potential conflicts of interest exist with any companies/organizations whose products or services may be discussed in this article.

Additional information: The e-Tables can be found in the Supplemental Materials section of the online article.

References

1. Rubin LJ. Introduction: diagnosis and management of pulmonary arterial hypertension: ACCP evidence-based clinical practice guidelines. *Chest*. 2004;126(1_suppl):7S-10S.
2. Benan M, Hande I, Gul O. The natural course of progressive systemic sclerosis patients with interstitial lung involvement. *Clin Rheumatol*. 2007;26(3):349-354.
3. White B. Interstitial lung disease in scleroderma. *Rheum Dis Clin North Am*. 2003;29(2):371-390.
4. Steen VD, Medsger TA. Changes in causes of death in systemic sclerosis, 1972-2002. *Ann Rheum Dis*. 2007;66(7):940-944.
5. Al-Dhaher FF, Pope JE, Ouimet JM. Determinants of morbidity and mortality of systemic sclerosis in Canada. *Semin Arthritis Rheum*. 2010;39(4):269-277.
6. Beretta L, Cappiello F, Barili M, Bertolotti F, Scorza R. T-889C IL-1alpha promoter polymorphism influences the response to oral cyclophosphamide in scleroderma patients with alveolitis. *Clin Rheumatol*. 2007;26(1):88-91.
7. Boin F, Wigley F, Wise R, Rosen A. Circulating T cell polarization is associated with respiratory decline in scleroderma patients with active interstitial lung disease [abstract]. *Arthritis Rheum*. 2010;62(suppl 10):1989.
8. Bouros D, Wells AU, Nicholson AG, et al. Histopathologic subsets of fibrosing alveolitis in patients with systemic sclerosis and their relationship to outcome. *Am J Respir Crit Care Med*. 2002;165(12):1581-1586.
9. de Lauretis A, Pantelidis P, Sestini P, et al. Serum markers of disease progression in idiopathic pulmonary fibrosis and in interstitial lung disease associated with systemic sclerosis. *Am J Respir Crit Care Med*. 2010;181:A1105.
10. De Santis M, Inzitari R, Bosello SL, et al. β-Thymosins and interstitial lung disease: study of a scleroderma cohort with a one-year follow-up. *Respir Res*. 2011;12:22.
11. De Santis M, Bosello SL, Peluso G, et al. Bronchoalveolar lavage fluid and progression of scleroderma interstitial lung disease. *Clin Respir J*. 2012;6(1):9-17.
12. Fertig N, Domsic RT, Rodriguez-Reyna T, et al. Anti-U11/U12 RNP antibodies in systemic sclerosis: a new serologic marker associated with pulmonary fibrosis. *Arthritis Rheum*. 2009;61(7):958-965.
13. Fischer A, Swigris JJ, Groshong SD, et al. Clinically significant interstitial lung disease in limited scleroderma: histopathology, clinical features, and survival. *Chest*. 2008;134(3):601-605.
14. Goh NS, Veeraraghavan S, Desai SR, et al. Bronchoalveolar lavage cellular profiles in patients with systemic sclerosis-associated interstitial lung disease are not predictive of disease progression. *Arthritis Rheum*. 2007;56(6):2005-2012.
15. Goh NS, Desai SR, Veeraraghavan S, et al. Interstitial lung disease in systemic sclerosis: a simple staging system. *Am J Respir Crit Care Med*. 2008;177(11):1248-1254.
16. Goh NS, Desai SR, Anagnostopoulos C, et al. Increased epithelial permeability in pulmonary fibrosis in relation to disease progression. *Eur Respir J*. 2011;38(1):184-190.
17. Khanna D, Tseng CH, Farmani N, et al. Clinical course of lung physiology in patients with scleroderma and interstitial lung disease: analysis of the Scleroderma Lung Study Placebo Group. *Arthritis Rheum*. 2011;63(10):3078-3085.
18. Kim EA, Johkoh T, Lee KS, et al. Interstitial pneumonia in progressive systemic sclerosis: serial high-resolution CT findings with functional correlation. *J Comput Assist Tomogr*. 2001;25(5):757-763.
19. Launay D, Remy-Jardin M, Michon-Pasturel U, et al. High resolution computed tomography in fibrosing alveolitis associated with systemic sclerosis. *J Rheumatol*. 2006;33(9):1789-1801.
20. Launay D, Humbert M, Berezne A, et al. Clinical characteristics and survival in systemic sclerosis-related pulmonary hypertension associated with interstitial lung disease. *Chest*. 2011;140(4):1016-1024.
21. Le Gouellec N, Faivre JB, Hachulla AL, et al. Prognosis factors for survival and progression-free survival in SSc-associated interstitial lung disease [abstract]. *Rheumatology*. 2012;51(suppl 2):ii82.
22. Lopes AJ, Capone D, Mogami R, Menezes SL, Guimarães FS, Levy RA. Systemic sclerosis-associated interstitial pneumonia: evaluation of pulmonary function over a five-year period. *J Bras Pneumol*. 2011;37(2):144-151.
23. Roth MD, Tseng CH, Clements PJ, et al; Scleroderma Lung Study Research Group. Predicting treatment outcomes and responder subsets in scleroderma-related interstitial lung disease. *Arthritis Rheum*. 2011;63(9):2797-2808.
24. Shahane A, Parambil J, Xu M, Chatterjee S. Scleroderma lung disease: effect of co-existent pulmonary hypertension on progression of interstitial lung disease [abstract]. *Arthritis Rheum*. 2010;62(suppl 10):592.
25. Steen VD, Conte C, Owens GR, Medsger TA Jr. Severe restrictive lung disease in systemic sclerosis. *Arthritis Rheum*. 1994;37(9):1283-1289.
26. Strange C, Bolster MB, Roth MD, et al; Scleroderma Lung Study Research Group. Bronchoalveolar lavage and response to cyclophosphamide in scleroderma interstitial lung disease. *Am J Respir Crit Care Med*. 2008;177(1):91-98.
27. Swigris JJ, Zhou X, Wamboldt FS, et al. Exercise peripheral oxygen saturation (SpO2) accurately reflects arterial oxygen saturation (SaO2) and predicts mortality in systemic sclerosis. *Thorax*. 2009;64(7):626-630.
28. Trad S, Amoura Z, Beigelman C, et al. Pulmonary arterial hypertension is a major mortality factor in diffuse systemic sclerosis, independent of interstitial lung disease. *Arthritis Rheum*. 2006;54(1):184-191.
29. Volpinari S, La Corte R, Bighi S, et al. Bronchoalveolar lavage in systemic sclerosis with lung involvement: role and correlations with functional, radiological and scintigraphic parameters. *Rheumatol Int*. 2011;31(9):1183-1188.
30. Zhang X, Bonner A, Hudson M, Baron M, Pope J. Association of gastroesophageal

factors and progression of interstitial lung disease in the Canadian Scleroderma Research Group (CSRG), a large, multi-center database. *J Rheumatol*. 2012;39(8):1707.

31. Kuwana M, Takeuchi T, Kaburaki J. Elevation of KL-6 at early disease course predicts subsequent deterioration of pulmonary function in patients with systemic sclerosis and interstitial lung disease [abstract]. *Arthritis Rheum*. 2012;64(suppl 10):850.
32. Tashkin DP, Elashoff R, Clements PJ, et al; Scleroderma Lung Study Research Group. Cyclophosphamide versus placebo in scleroderma lung disease. *N Engl J Med*. 2006;354(25):2655-2666.
33. Ley B, Ryerson CJ, Vittinghoff E, et al. A multidimensional index and staging system for idiopathic pulmonary fibrosis. *Ann Intern Med*. 2012;156(10):684-691.
34. du Bois RM, Weycker D, Albera C, et al. Ascertainment of individual risk of mortality for patients with idiopathic pulmonary fibrosis. *Am J Respir Crit Care Med*. 2011;184(4):459-466.
35. Wells AU, Desai SR, Rubens MB, et al. Idiopathic pulmonary fibrosis: a composite physiologic index derived from disease extent observed by computed tomography. *Am J Respir Crit Care Med*. 2003;167(7):962-969.
36. Kim EJ, Elicker BM, Maldonado F, et al. Usual interstitial pneumonia in rheumatoid arthritis-associated interstitial lung disease. *Eur Respir J*. 2010;35(6):1322-1328.
37. Ryerson CJ, Urbania TH, Richeldi L, et al. Prevalence and prognosis of unclassifiable interstitial lung disease. *Eur Respir J*. 2013;42(3):750-757.
38. Nicholson AG, Fulford LG, Colby TV, du Bois RM, Hansell DM, Wells AU. The relationship between individual histologic features and disease progression in idiopathic pulmonary fibrosis. *Am J Respir Crit Care Med*. 2002;166(2):173-177.